

**This assignment will not be graded and is only for practice.**

**Key Points: 1**

Two non-zero vectors  $u$  and  $v$  from an inner product space  $V$  are said to be orthogonal if  $\langle u, v \rangle = 0$ .

A set  $S$  of non-zero vectors is said to be an orthogonal set of an inner product space if elements of the set  $S$  are mutually orthogonal. i.e.,  $\langle v_i, v_j \rangle = 0$  for  $i \neq j$ , any  $v_i, v_j \in S$ .

Let  $\{v_1, v_2, \dots, v_n\}$  be an orthogonal set of vectors in an inner product space. Then  $\{v_1, v_2, \dots, v_n\}$  is a linearly independent set of vectors.

**Orthogonal basis:** Let  $V$  be an inner product space. A basis consisting of mutually orthogonal vectors is called an orthogonal basis. In other word, a maximal orthogonal set is a basis and is called an orthogonal basis.

An orthonormal set of vectors of an inner product space  $V$  is an orthogonal set of vectors such that the norm of each vector of the set is 1.

An orthonormal basis is an orthonormal set of vectors which form a basis.

**Obtaining orthonormal set from orthogonal set:** If  $\gamma = \{v_1, v_2, \dots, v_n\}$  is an orthogonal set of vectors then we can obtain an orthonormal set of vectors  $\beta$  from  $\gamma$  by dividing each vector by its norm i.e.,

$$\beta = \left\{ \frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}, \dots, \frac{v_n}{\|v_n\|} \right\}.$$

Consider a set of vectors  $S =$

$\{(1, -1, -1), (1, 2, 0), (-1, 1, 1), (-1, \frac{1}{2}, -\frac{3}{2}), (0, 3, 1), (2, 1, 1), (0, 1, -1)\}$  from the inner product space  $\mathbb{R}^3$  with usual inner product (i.e., the dot product).

Answer questions (1) and (2).

1) Which of the following subsets of the set  $S$  consist of mutually orthogonal vectors? **1 point**

- ☐  $\{(1, -1, -1), (1, 2, 0), (-1, 1, 1)\}$
- ☐  $\{(1, -1, -1), (-1, \frac{1}{2}, -\frac{3}{2})\}$
- ☐  $\{(1, -1, -1), (-1, \frac{1}{2}, -\frac{3}{2}), (1, 2, 0)\}$
- ☐  $S$
- ☐  $\{(2, 1, 1), (1, -1, -1), (0, 1, -1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\{(1, -1, -1), (-1, \frac{1}{2}, -\frac{3}{2})\}$$

$$\{(2, 1, 1), (1, -1, -1), (0, 1, -1)\}$$

2) Which of the following option(s) is (are) true?

**1 point**

- ☐  $\{(2, 1, 1), (0, 1, -1)\}$  is an orthogonal basis of a vector space which is isomorphic to  $\mathbb{R}^2$ .
- ☐  $\{(2, 1, 1)\}$  is not an orthogonal basis of a vector space which is isomorphic to  $\mathbb{R}$ .
- ☐  $\{(1, -1, -1), (1, 2, 0)\}$  is an orthogonal basis of a vector space which is isomorphic to  $\mathbb{R}^2$ .
- ☐  $\{(-1, 1, 1), (-1, \frac{1}{2}, -\frac{3}{2}), (0, 3, 1), \}$  is an orthogonal basis of  $\mathbb{R}^3$ .
- ☐  $\{\frac{(2,1,1)}{\sqrt{6}}, \frac{(1,-1,-1)}{\sqrt{3}}, \frac{(0,1,-1)}{\sqrt{2}}\}$  is an orthonormal basis of  $\mathbb{R}^3$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\{(2, 1, 1), (0, 1, -1)\} \text{ is an orthogonal basis of a vector space which is isomorphic to } \mathbb{R}^2.$$

$\left\{ \frac{(2,1,1)}{\sqrt{6}}, \frac{(1,-1,-1)}{\sqrt{3}}, \frac{(0,1,-1)}{\sqrt{2}} \right\}$  is an orthonormal basis of  $\mathbb{R}^3$ .

### Key Points: 2

**The projection of a vector to a subspace:** Let  $V$  be an inner product space,  $v \in V$  and  $W \subseteq V$  be a subspace. Then the projection of  $v$  onto  $W$  is the vector in  $W$ , denoted by  $\text{proj}_W(v)$ , computed as follows:

Find an orthonormal basis  $\{v_1, v_2, \dots, v_n\}$  for  $W$ .

Then  $\text{proj}_W(v) = \sum_{i=1}^n \langle v, v_i \rangle v_i$ .

Remark:  $\text{proj}_W(v)$  is independent of the chosen orthonormal basis.

Let  $V$  be an inner product space and  $W$  be a subspace. Then the projection of vectors in  $V$  to  $W$  is a linear transformation from  $V$  to  $V$  with image  $W$ .

3) Consider the vector space  $\mathbb{R}^3$  with usual inner product. Which of the following vectors represents projection of the vector  $u = (2, 1, -2)$  onto the subspace whose an orthonormal basis is  $\{(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}), (\frac{-2}{3}, \frac{1}{3}, \frac{2}{3})\}$ ? **1 point**

☐  $(\frac{14}{9}, \frac{-7}{9}, \frac{-14}{9})$

☐  $(1, 1, 1)$

☐  $(2, 1, -2)$

☐  $(\sqrt{2}, 0, \sqrt{2})$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(\frac{14}{9}, \frac{-7}{9}, \frac{-14}{9})$

4) Consider the vector space  $\mathbb{R}^3$  with usual inner product. If  $(a, b, c)$  is the projection vector obtained from projecting the vector  $(1, 0, 1)$  onto the subspace  $\{(x, y, z) \mid x - y + z = 0, x, y, z \in \mathbb{R}\}$ , then find the value  $3(a + b + c)$ .

**Hint:** Find any orthonormal basis of the subspace  $\{(x, y, z) \mid x - y + z = 0, x, y, z \in \mathbb{R}\}$  and use the definition of the projection of a vector onto a subspace.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 4

**1 point**

**Key Points: 3**

**Gram-schmidt process:** The Gram-schmidt process is used to get an orthogonal basis from any given basis of a vector space. A slight modification yields an orthonormal basis.

Let  $\gamma = \{v_1, v_2, \dots, v_n\}$  be a given ordered basis of a vector space. The steps in the process are described below and yield an orthonormal basis  $\beta = \{u_1, u_2, \dots, u_n\}$ .

Step-1  $w_1 = v_1$ , and  $u_1 = \frac{w_1}{\|w_1\|}$ .

Step-2  $w_2 = v_2 - \langle v_2, u_1 \rangle u_1$  and  $u_2 = \frac{w_2}{\|w_2\|}$ .

Step-3  $w_3 = v_3 - \langle v_3, u_1 \rangle u_1 - \langle v_3, u_2 \rangle u_2$  and  $u_3 = \frac{w_3}{\|w_3\|}$ .

$\vdots$

Step-i  $w_i = v_i - \sum_{j=1}^{i-1} \langle v_i, u_j \rangle u_j$  and  $u_i = \frac{w_i}{\|w_i\|}$ .

$\vdots$

Step-n  $w_n = v_n - \sum_{j=1}^{n-1} \langle v_n, u_j \rangle u_j$  and  $u_n = \frac{w_n}{\|w_n\|}$ .

Let  $\beta$  be an ordered basis of the inner product space  $\mathbb{R}^3$  with usual inner product, where  $\beta = \{(1, 1, -1), (1, 1, 0), (1, 0, 0)\}$  and  $\gamma = \{u_1, u_2, u_3\}$  be the corresponding orthonormal basis obtained using the Gram-schmidt process. Use this information to answer the questions (5), (6) and (7).

5) If  $u_1 = (a_1, b_1, c_1)$ , then find the value of  $\sqrt{3}(a_1 + b_1 + c_1)$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

**1 point**

6) If  $u_2 = (a_2, b_2, c_2)$ , then find the value of  $\sqrt{6}(a_2 + b_2 + c_2)$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 4

**1 point**

7) If  $u_3 = (a_3, b_3, c_3)$ , then find the value of  $\frac{1}{\sqrt{2}}(a_3 + b_3 + c_3)$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

**1 point**

### Key Points: 6

**Orthogonal transformation:** Let  $V$  be an inner product space and  $T$  be a linear transformation from  $V$  to  $V$ .  $T$  is said to be an orthogonal transformation if

$$\langle Tv, Tw \rangle = \langle v, w \rangle, \forall v, w \in V$$

**Orthogonal matrix:** A square matrix  $A$  is called an orthogonal matrix if  $AA^T = A^T A = I$  (identity matrix)



8) Consider the following linear transformations  $T_1, T_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ , and  $T_3, T_4 : \mathbb{R}^3 \rightarrow \mathbb{R}^2$  defined as **1 point**

$$T_1(x, y) = \left( \frac{x+y}{\sqrt{2}}, \frac{x-y}{\sqrt{2}} \right),$$

$$T_2(x, y) = (x + 3y, 2x - 2y),$$

$$T_3(x, y, z) = \left( \frac{x+z}{\sqrt{2}}, \frac{x+y\sqrt{3}+z}{\sqrt{3}} \right),$$

$$T_4(x, y, z) = (x + y + z, x + y),$$

where  $\mathbb{R}^3$  and  $\mathbb{R}^2$  are the inner product spaces with respect to the dot product. Which of the linear transformation is an orthogonal transformation?

☐  $T_1$

☐  $T_2$

☐  $T_3$

☐  $T_4$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T_1$

9) Which of the following option(s) is (are) true?

**1 point**

☐  $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$  is an orthogonal matrix.

☐  $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$  is an orthogonal matrix.

☐  $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 2 \\ -1 & -2 & 1 \end{bmatrix}$  is an orthogonal matrix

☐  $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$  is an orthogonal matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$  is an orthogonal matrix.

$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$  is an orthogonal matrix.

$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}$  is an orthogonal matrix.

Your score is: 0/9